SaaqibMart — E-Commerce Web Application Report

Module: B204 App & Web Development

Institution: Gisma University of Applied Sciences, School of Computer Science

Assessment: Individual Project (Reassessment)

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GitHub Repository: https://github.com/Saaqibx77/B204WebApplicationSaaqibMart

1. Introduction

This project is a simple e commerce web application designed for online mobile sales. It is developed as part of B204 App & Web Development Module to demonstrate skills in full stack web development with the help of Technology HTML,CSS, JavaScript, NodeJs, Json, expressJs.

Front end is designed using HTML which provides structure to the web site, CSS Provides the Look and feel and simple User interface which is easy to use. For Data Storage in the Back End related to all orders made Nodejs, ExpressJs is used with data storage is stored in orders.json file where data is persistence also proper system is there for complete payment integration and purchase workflow.

The application helps the end users to search a catalogue of smartphones, add selected items to a shopping cart, proceed through a multi-step checkout with shipping and payment details, and place orders that are stored in the json file for the future reference.

This report covers all the desin process, implantation process and all Technical aspects that are behind the SaaqibMart Web application. In the application I have designed the front end with the Help of HTML, CSS with that Back end is connected it has very good back end architecture, complete database design,Complete order system where we can select the items, select the quantity give the customer information give the payment information , card details, paypal details etc.

2. Project Overview and Objectives

It is a simple and easy to use single page e commerce web application.

Website provides the complete handling of the online shopping and understands the complete work flow of the website which is required in a mobile sale web application.

The main objectives of the project are:

1. **Easy to Use Landing Page:**
   Easy to use landing page which has simple option to select the products, make orders and the all other information that is required for easy functioning of the web site.
2. **List of all the Products:**
   Web application display all available smartphones in easy to use responsive web page and simple grid format to use it, prices are easily visible in Euros by which user can add any product by selecting "Add to Cart" buttons which is available on every grid element.
3. **Shopping Cart:** web application provides a simple side bar where we change the quantity, ,delete the item and can see individual total and the Total cost and given a Proceed to Checkout Button
4. **Searching option :** It allows searh out the product by name if not clearly visibile.
5. **Payment Integration:**
   Project support and shows the complete payment process of selecting the payment option, enter card details or paypal details to make order.
6. **Orders Storage:**
   All the placed orders are stored permanently with the help of NodeJs in the Json file orders.json
7. **Responsive Design:**
   The application can be easily used on computer, mobiles or tablets on all platforms.

3. User Interface Design (Front-End)

3.1 Design Philosophy

The visual design of my project follows a simple dark theme which is easy to use and simple with a gold accent colour, it provides a good and decent look good for a mobile retail store.

The font generally preferred and referred from Google Fonts with the weights which is easily visible to the end users in the range 300 to 700.

3.2 Landing Page

Landing page Fulfils the basic requirements which is essential for “an attractive and easy to use landing page” it consists of 3 parts

- **Hero Section:** It consists a banner which spread to complete page with simple tagline "Your Next Phone Is Here” In this section a dark background is provided with the help of CSS.
- **About Section:** Titled "Why SaaqibMart", this section tells brief about the company and communicates company's value proposition
- **Contact Section:** A simple Contact form which consists of name, message and email which allow user to submit inquiries and a notification by using toast.

3.3 Product Listing Page

It displays all the available smartphones with a simple CSS Grid it uses responsive Grid template

A product card is designed with product name, product price in euro a simple image and an Add to Cart Button. Grid automatically to show number of columns based on the width of the viewport. Add to cart button is available on clicking that button item is added to the cart.

3.4 Navigation and Accessibility

The navigation bar is given at the top with a simple logo navigation buttons are provided with navigation link provided Home, Products, About and Contact, also a cart button is provided which shows a badge with live count badge. In addition, a Hamburger menu is provided for mobile viewports.

Easy accessibility is provided with HTML 5 semantics that includes nav for navigation bar, header, section, aside bar.

4. Front-End Technologies and Implementation

4.1 HTML Structure

The application consists of a single index.html file, which contains all the page sections the cart sidebar, which is easy to use, and a checkout modal and using toast for notification.

The HTML uses all the HTML 5 semantic elements in the website development: nav for the navigation bar, header is used for the hero section, also section is provided for content blocks, aside is used for the cart sidebar, and footer is used for the page footer. Meta tag is used for providing charset information.

4.2 CSS Styling

The style.css file provide the complete design for the web site, design and color combination is provided with the hex code that provides the implementation with minimal design following are some of the CSS techniques which are used in project development.

- **CSS Grid is used** for the product card layout with auto-fill and minmax() for responsive columns.
- **Flexbox:** Flexbox is used for designing navbar, cart items, rows of checkout and payment method flexbox.
- **CSS Transitions:** Some Transitions are used to the cart sidebar and toast notification for functional animations.

4.3 JavaScript Application Logic

The script.js file handles all front end logic. The code is provided with following modules

- **Product Showing:** The renderProducts() function in javascript is used to generate product cards which gets it data from products array and display them in the front end.
- **Cart Management:** Functions are used in JavaScript for adding products to cart addToCart(), Function to increase or decrease Cart Items count changeQty() function is used, removeFromCart() is used to delete items from the cart
- **Checkout Flow:** Check out flow is controlled by 3 functions openCheckout(), showStep(), goToPayment(), backToShipping(), and placeOrder() used to handle Shipping, payment and confirmation.
- **Changing Payment Method :** The selectPayment() function is used to shows/hides form fields based on the selected payment method (Card, PayPal, or Bank Transfer).
- **Helper Functions:** Helper functions are used for Euro formatting (eur()), toast notifications (showToast()), function for card number formatting (formatCard())

5. Back-End Development

5.1 Server Architecture

The back end is implemented in server.js using the code in Node.js with the support of Express.js Framework. Server Listens to the port 3000 and uses express.json middleware to parse incoming JSON data.

```
const express = require("express");
const app = express();
app.use(express.json());
app.use(express.static(__dirname));
app.listen(3000);
```

5.2 Database Schema

Simple JSON file (orders.json) used as a lightweight document store for order transactions permanent data. As our coursework B204 brief recommends MongoDB, approach of the JSON file was chosen for simplicity and easy deployment with no dependency.

6. Payment Integration

In the web application payment process supporting three payment methods that includes Credit or Debit card, Paypal or Bank transfer.

6.1 Credit/Debit Card payment methods

When the user selects the Card payment method 4 digit inputs will be displayed, automatic formatting is used to insert spaces after every 4 digits. Expiry Date is formatted to "MM / YY,

CVV (maxlength of 4 digits), and Cardholder Name. all the fields are mandatory to be entered for that HTML5 required attribute is used.

6.2 PayPal

When the Paypal option is selected in the payment option than cardfields will be hidden and will be replaced by single Paypal Email Address.

6.3 Bank Transfer

When Bank transfer option is selected a static type panel will be displayed in which user to include their order number which is dynamically generated by the system, order number will be referred as payment reference number.

6.4 Transaction Confirmation

After placing the order the web application displays a confirmation screen with a green checkmark icon that confirms that payment is made though in the current system simulates the payment method so for the security reasons payment is made virtually and actual account will be debited or credited. A automatic unique order number is generated with the prefix MM- that can help to be used to maintain order details. Also the details will be stored in the orders.json in the json format for future reference.

7. Purchasing Process working

The complete purchasing process in the web application is described below.

a) **Browse Products:** the user scrolsThe user scrolls to the "Top Selling Phones" section and views all smartphone available with model names and prices in euro.
b) **Add to Cart:** Add to cart button is available in User interface. With the help of add to cart option user can select the desired product add to card that will be visible in top right side of the window cart badge is automatically updated
c) **Review Cart:** user can click on the cart icon to open the side bar. The cart will be displayed with item and an icon, name and the unit price for quantity control (+/-) buttons are provided and a total will be visible The cart displays each item with its emoji, name, unit price, quantity controls (+/- buttons), a remove button, and a running total at the bottom.
d) **Adjust Quantities:** The user can increase or decrease the quantities easily using (+/-) button a delete icon is also available by which we can delete the required product from the cart.
e) **Proceed to Checkout:** Proceed to check out button is available, user clicks the proceed to check out button user will be redirected to the shipping details screen.
f) **Enter Shipping Details:** In this window user has to enter the shipping details that includes full name, email address, address , city, postal code and country all the data will be validated after further processing web application will be redirected to select payment method page.
g) **Select Payment Method:** after selecting the "Continue to Payment," the user can see the order summary user can select one of the payment method out of three payment methods: Credit/Debit Card, PayPal, or Bank Transfer. The form than display the required details and redirected to the place order option
h) **Place Order:** The user clicks on "Place Order." The front-end sends a POST request to /api/orders with the complete order data of the purchased item. The server saves the order details in the orders.json with a timestamp after this a order will be recorded and order confirmation message will be displayed.
i) **Order Confirmation:** The user will see a confirmation screen with a green checkmark, a success message, and the unique order number will be displayed that web application will generate. The cart is cleared again and form is reset.

8. Setup Instructions

8.1 Prerequisites

- Node.js version 14 or higher installed on the system.

8.2 Installation and Running

```
# 1. Clone the repository
git clone https://github.com/yourusername/SaaqibMart.git
cd SaaqibMart

# 2. Install dependencies
npm install

# 3. Start the server
npm start

# 4. Open in browser
http://localhost:3000
```

8.3 Project File Structure

```
SaaqibMart/
├── index.html        # Main HTML page (all sections + cart + checkout)
├── style.css         # Minimal stylesheet
├── script.js         # Front-end logic (cart, checkout, API calls)
├── server.js         # Express server (static files + order API)
├── package.json      # Node.js dependencies and scripts
├── orders.json       # Auto-created: stores all placed orders
└── README.md         # Project documentation
```

9. Testing and Quality Assurance

The application was tested for different inputs with the required data and checked for the quality and data storage in the back end in the orders.json file.

- **Cart Functionality:** cart functionality is tested by adding the items in the cart. Adjusting quantities increase or decrease items also user can remove items if not required. For each operation system will be tested.
- **Checkout Flow:** Tested the complete three step checkout process which will be tested for adding items, selecting shipping method, selecting payment method and make payment.
- **Cross-Browser Testing:** The application is tested to run in Google Chrome, Mozilla Firefox, and Microsoft Edge. All features are working perfectly on all the browsers.
- **Responsive Testing:** the web application is tested for different size of screen like tested for mobile, tablet, laptop and desktop for different pixel size like (375px, 768px, 1024px, 144px) Responsive layout work perfectly for all type of layouts.

10. Conclusion

The web application successfully demonstrates the application of full-stack web development skills that we have learned in the B204 module. The project covers the fully functional ecommerce website requirements. It is visually very clean and easy to understand. The data for all the orders is properly stored in the orders.json file with the help of Node.js and Express.js back-end .

Important technical parts that is covered is to provide a fully functional css design without any dependency on any framework.

a clean JavaScript architecture is used to provide the complete functionality which is required in the project.

Areas for future enhancement include migrating from current JSON file storage system to MongoDB or sql based or sqlite based storage system for real life usage.

Provide complete CURD operations with stock storage, records of all the customers creating login for the users so that all visitors and customer records will be maintained. AI chatbot and AI integration for AI based search and other AI based facilities.

11. References

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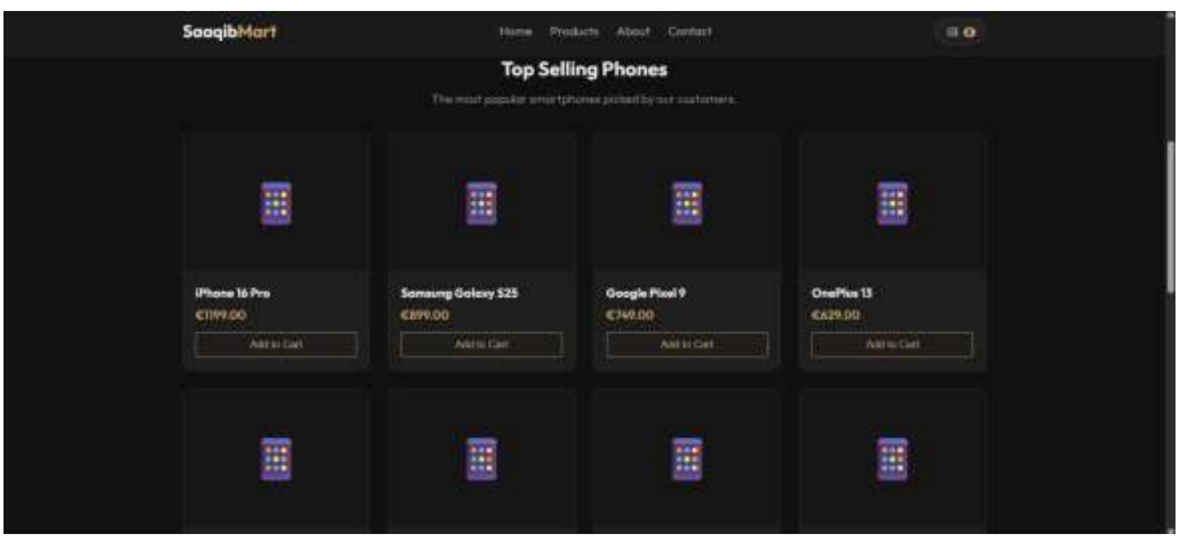
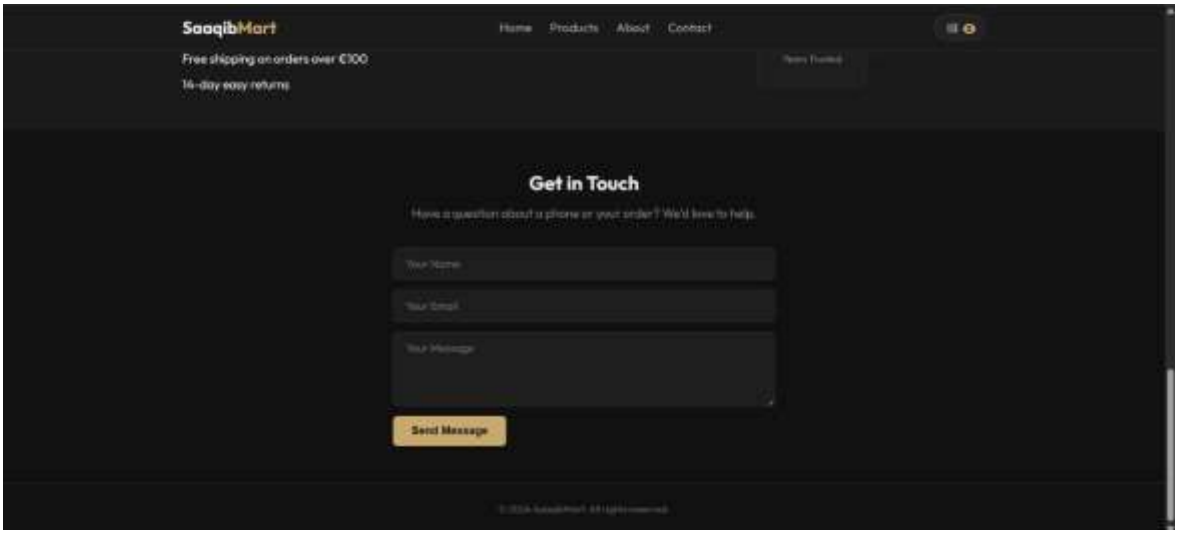
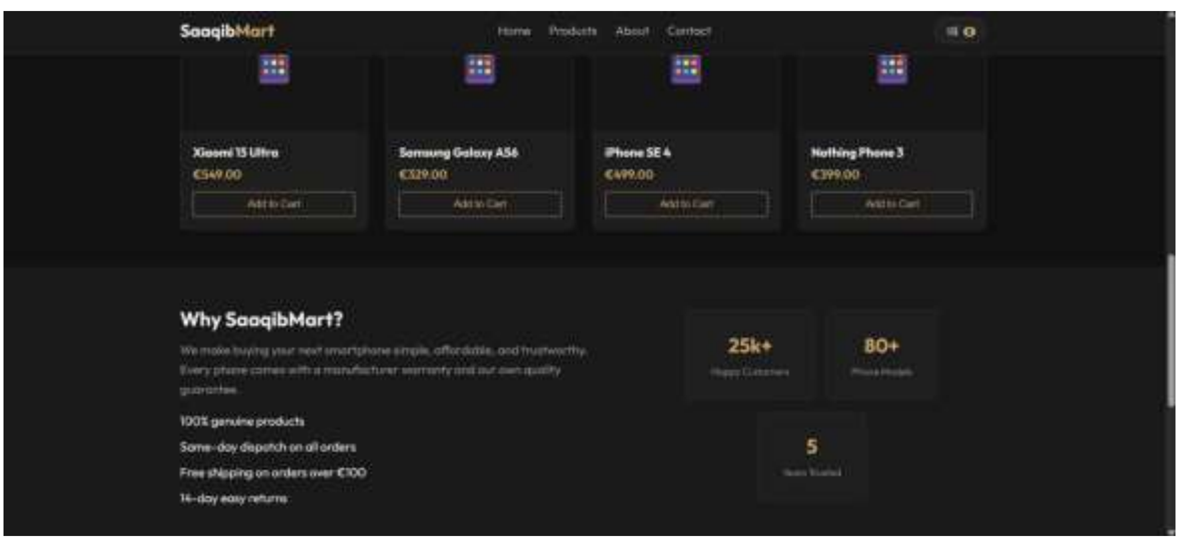
PayPal Developer (2026) PayPal Sandbox Testing Guide. Available at: https://developer.paypal.com/docs/api-basics/sandbox/ (Accessed: 1 July 2026).

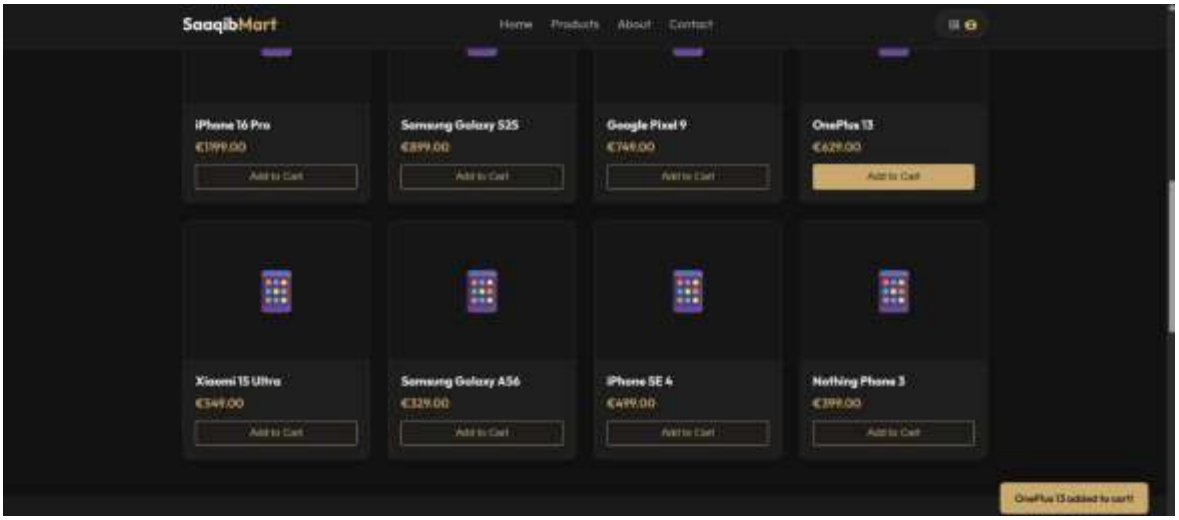
Appendix


SaaqibMart
Home
Products
About
Contact
LATEST SMARTPHONES
Your Next Phone Is Here
Top brands, best prices, and fast delivery — find the perfect smartphone for you.
Browse Phones
Top Selling Phones
The most popular smartphones picked by our customers.

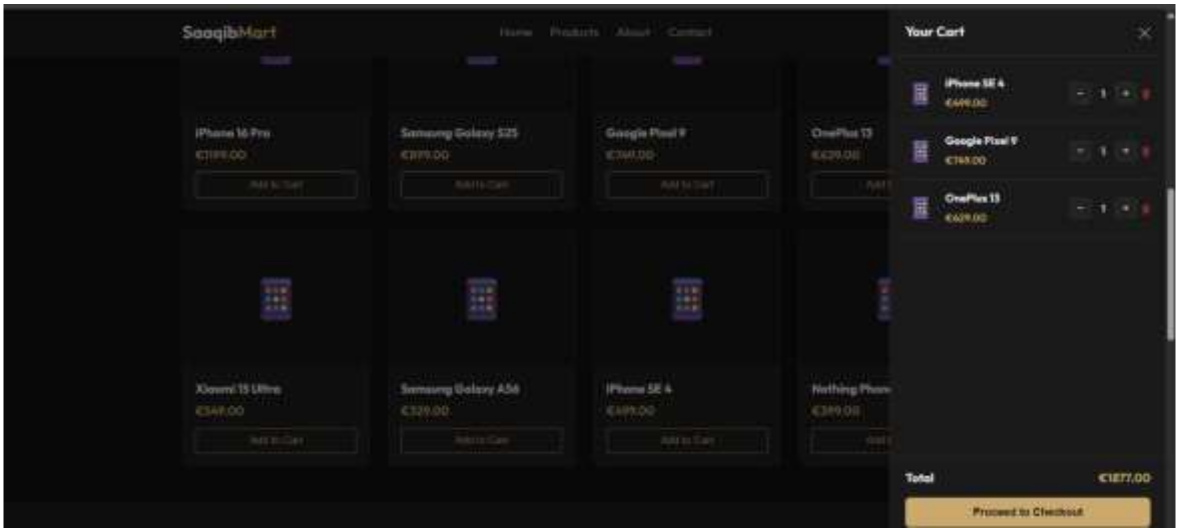

SaaqibMart
Home
Products
About
Contact
Top Selling Phones
The most popular smartphones picked by our customers.
iPhone 16 Pro
€1199.00
Add to Cart
Samsung Galaxy S25
€899.00
Add to Cart
Google Pixel 9
€749.00
Add to Cart
OnePlus 13
€629.00
Add to Cart

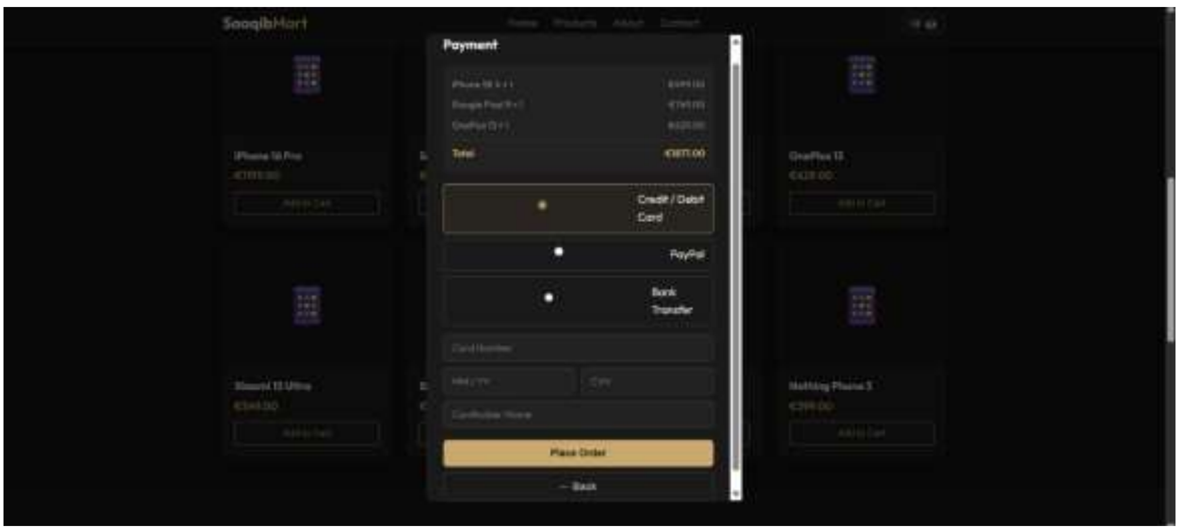

SaaqibMart
Home
Products
About
Contact
Free shipping on orders over €100
14-day easy returns
Get in Touch
Have a question about a phone or your order? We'd love to help.
Your Name
Your Email
Your Message
Send Message

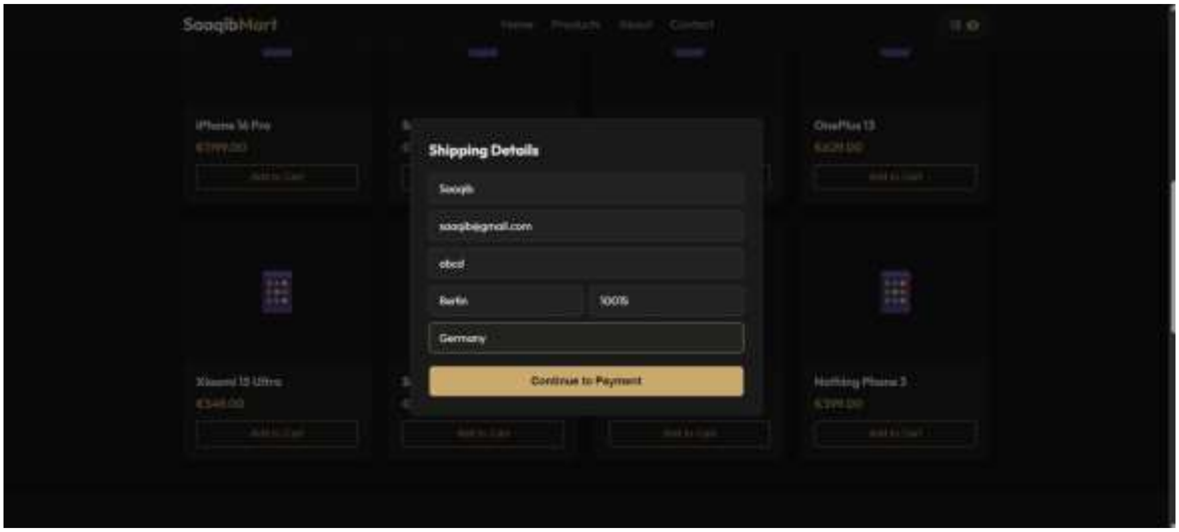

SaaqibMart
Home
Products
About
Contact
Xiaomi 15 Ultra
€549.00
Add to Cart
Samsung Galaxy A56
€329.00
Add to Cart
iPhone SE 4
€499.00
Add to Cart
Nothing Phone 3
€399.00
Add to Cart
Why SaaqibMart?
We make buying your next smartphone simple, affordable, and trustworthy. Every phone comes with a manufacturer warranty and our own quality guarantee.
100% genuine products
Same-day dispatch on all orders
Free shipping on orders over €100
14-day easy returns
25k+
Happy Customers
80+
Phone Models
5
Years Trusted

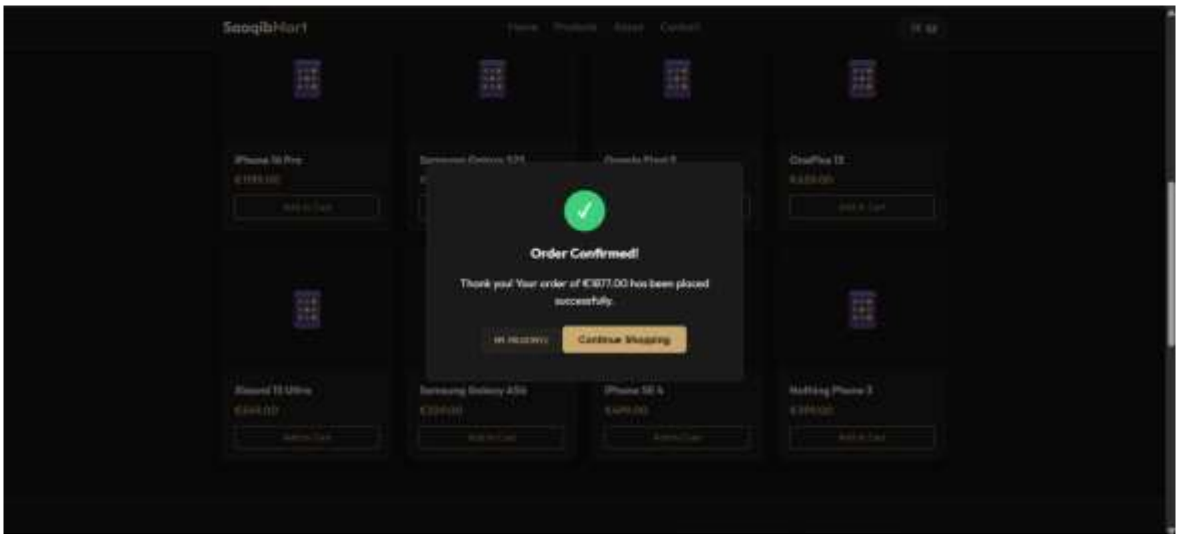


SaaqibMart
Home
Products
About
Contact
iPhone 16 Pro
€1199.00
Add to Cart
Samsung Galaxy S25
€899.00
Add to Cart
Google Pixel 9
€749.00
Add to Cart
OnePlus 13
€629.00
Add to Cart
Xiaomi 15 Ultra
€549.00
Add to Cart
Samsung Galaxy A56
€329.00
Add to Cart
iPhone SE 4
€499.00
Add to Cart
Nothing Phone 3
€399.00
Add to Cart
OnePlus 13 added to cart!


SaaqibMart
Home
Products
About
Contact
Your Cart
iPhone 16 Pro
Samsung Galaxy S25
Google Pixel 9
OnePlus 13
Xiaomi 15 Ultra
Samsung Galaxy A56
iPhone SE 4
Total
Proceed to Checkout


SaaqibMart
Payment
Total
Credit / Debit Card
PayPal
Bank Transfer
Place Order
← Back


SaaqibMart
Shipping Details
Saaqib
saaqib@gmail.com
abcd
Berlin
10015
Germany
Continue to Payment


SaaqibMart
Order Confirmed!
Thank you! Your order of €1877.00 has been placed successfully.
Continue Shopping

orders.json

orders.json > ...


```json
[
  {
    "orderNum": "MM-TEST123",
    "total": 1199,
    "items": [
      {
        "id": 1,
        "name": "iPhone 16 Pro",
        "price": 1199,
        "qty": 1,
        "subtotal": 1199
      }
    ],
    "shipping": {
      "name": "Test User",
      "email": "test@test.com",
      "address": "123 Test St",
      "city": "Berlin",
      "zip": "10115",
      "country": "Germany"
    },
    "paymentMethod": "card",
    "createdAt": "2026-07-03T03:23:33.510Z"
  }
]
```